1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

TLS-like lymphoid aggregates confer sensitivity to gemcitabine plus PD-L1 blockade only when the tumor microenvironment supports a robust IFNG-driven transcriptional program that upregulates antigen presentation (MHC I/II) and strengthens CD8+ T cell cytotoxic activity. Consequently, TLS-rich tumors with high IFNG signaling will exhibit higher therapy_response to gemcitabine + anti-PD-L1 than TLS-rich tumors with low IFNG signaling. In parallel, intrinsic invasion programs and an immunosuppressive myeloid milieu can dampen or override IFNG-driven responses, leading to early relapse despite an initial response. Therefore, a biologically defined responder subset is determined by (i) TLS presence and maturation, (ii) sufficient IFNG activity, and (iii) competent antigen presentation, whereas high-grade/invasive phenotypes and hostile myeloid environments set a threshold for durability. The hypothesis implies testable predictions: (a) among TLS-rich tumors, therapy_response correlates positively with an IFNG activity score and MHC I/II expression; (b) disrupting IFNG signaling abrogates the TLS-dependent response; (c) enhancing IFNG signaling or antigen presentation in TLS-rich, IFNG-low tumors converts them toward sensitivity; (d) invasion-program signatures predict early relapse even when initial response is observed; (e) PD-L1 expression alone does not predict response without concurrent IFNG-driven antigen presentation.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	IFNG
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

IFNG: IFNG-mediated immune activation, yet invasive and tertiary higher-grade tumor features drive recurrence and metastasis — so why do some tumors with apparent immune architecture fail to respond to the combination? 2) Tier 1 paths used: (a) the gemcitabine + Ate-Grab → IFNG axis (supports a mechanism whereby combination therapy relies on induction of an IFNG program to enable anti-tumor immunity); (b) the invasion/AdCC path and the tertiary higher-grade component → recurrence link (highlights tumor-intrinsic invasive/aggressive programs that can undermine immune control).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PD-L1 (CD274)	Supporting modifier from knowledge graph

Secondary	MHC class I/II	Supporting modifier from knowledge graph
Secondary	MMPs (matrix metalloproteinases)	Supporting modifier from knowledge graph
Secondary	tertiary higher-grade component	Supporting modifier from knowledge graph
Secondary	myeloid markers (CD11b, CD33)	Supporting modifier from knowledge graph
Secondary	lymphoid markers (CD3, CD8)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- PD-L1 (CD274): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MHC class I/II: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.